



DOCUMENTATION

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INTRO

For my Development 5 task, I chose to create a Zombie Idle Shooter game called ZombieFarm. I integrated Firebase to handle the real-time database and player data storage.

Initially, I considered using SQL for data management, but Firebase proved to be a better fit for my needs—offering faster, more reliable performance especially for Android builds. This project marks my first-ever Android game build.

I developed the game in an agile, sprint-based manner, focusing on Minimum Viable Product (MVP) increments. By coding small portions iteratively, I gradually built a sizable codebase.

To assist my development, I leveraged AI for hints, problem-solving, and generating larger sections of code—especially for bonus features like the leaderboard and Android touch input support. While I used AI-generated content for these, I made personal modifications so the features aligned perfectly with my vision.

The core game functionality resides in the GameManager script, where most of the gameplay logic and interactions between different modules are managed.

Throughout development, I also drew many sketches and notes on paper, helping to organize ideas and define the MVP and scope.

The game has been thoroughly tested on both Android and PC platforms, working seamlessly on both. Data storage functions correctly, and the overall experience is stable and ready.

In the next section, I will explain the game overview.

Enjoy reading!

Game Overview

ZombieFarm is an engaging idle shooter game where players defend their farm against relentless waves of zombies. Using taps and strategically placed survivors, players survive escalating waves, defeat powerful bosses, and climb a global leaderboard. The game features persistent online saves via Firebase, making every session count.

Core Gameplay Loop

- Shop
 - Open Shop => Select Lane => Buy/Sell survivor => Buy PowerUps
 - Face challenging boss waves that require rapid tapping.
 - Earn bonuses and progress to higher waves.
 - Track stats and compete in global leaderboards.
-

How to Play

1. Register or Login

- Login/Register Screen, Choose a Username and Password so that your progress is saved

2. Buy Survivors

Open Up the shop with the top left button, select a lane then Buy/ or sell Survivor,

- Soldier costs 20€, Commando costs 40€ and General costs 60€
- Soldier => Regular pace, 1 damage, Commando => Fast Pace, 1 damage => General => Intermediate Pace, 2 Damage



3. Upgrade Survivors in the Shop

- Open Shop, and Press and choose powerup, speed or damage boost, to finish waves faster



4. Face the Zombie Boss

- After a wave ends, prepare yourself to tap as much as you can on the ZombieBoss to add bonus money



5. Check the Leaderboard

- Press Leaderboard button on top left corner, under the Open Shop to see the leaderboard and your current rank, you can return to game or leave the game



4. Face the Zombie Boss

- After a wave ends, prepare yourself to tap as much as you can on the ZombieBoss to add bonus money



SETUP INSTRUCTIONS

This section covers all the necessary prerequisites, project setup steps, Firebase integration, and building your APK for Android.

A. Requirements

Before starting, make sure you have the following installed and configured:

- Unity 6 (6000.0.25f1) – The version used for development and testing.
- Unity Hub – Recommended for managing Unity versions and projects.
- Firebase Unity SDK – Includes Authentication and Realtime Database packages.
- TextMeshPro (TMP) – For UI text components.
- Android SDK & NDK – Required for Android builds (via Unity's Android Build Support).
- Internet Connection – Needed to access Firebase services during testing.

B. Unity Project Setup

Clone the Repository

Open your terminal or command prompt, then run:

```
git clone https://github.com/your-username/zombiefarm.git
```

Open the Project

- Launch Unity Hub.
- Click Open, and select the cloned zombiefarm project folder.
- Wait for Unity to load and import all assets and packages.

Install Required Packages

If not pre-imported, open Window > Package Manager, and:

- Locate and install TextMeshPro.
- Import the Firebase Unity SDK .unitypackage files if not included already.

SETUP INSTRUCTIONS

C. Firebase Setup

1. Create Firebase Project
 - Visit [Firebase Console](#).
 - Create a new project for ZombieFarm.
2. Add Android App
 - In your Firebase project, select Add App > Android.
 - Register your app with the correct package name (e.g., com.yourname.zombiefarm).
 - Download the google-services.json file.
3. Place google-services.json
 - Copy the downloaded google-services.json into your Unity project under the Assets folder.
4. Enable Firebase Services
 - In Firebase Console, go to Authentication and enable Email/Password sign-in.
 - Go to Realtime Database, create the database, and set rules:

Example rules for development:

```
json
{
  "rules": {
    ".read": true,
    ".write": true
  }
}
```

(Remember to tighten these rules before production.)

1. Import Firebase SDK in Unity
 - Import the Firebase Auth and Realtime Database SDK packages into your Unity project.
 - Confirm the Firebase initialization in your FirebaseAuthManager script.

SETUP INSTRUCTIONS

D. Building the APK (Android)

1. Open File > Build Settings.
2. Switch the platform to Android.
3. Verify all required scenes (ZombieFarm, LeaderBoard) are listed in the Scenes In Build.
4. Connect an Android device or setup an emulator.
5. Click Build And Run to compile and deploy the APK directly.
6. Alternatively, click Build to export the APK file to distribute manually (e.g., upload to Itch.io).

Notes

- Ensure your device allows installation from unknown sources if side-loading APKs for testing.
- Use the Unity console for build errors or warnings and consult Firebase logs for authentication issues.
- For detailed Firebase configuration, see Documentation/FirebaseSetup.md.

CODE ARCHITECTURE

A. SCRIPT OVERVIEW

Script	Role
AuthUIController.cs	Handles registration/login UI and Firebase
GameManager.cs	Central game logic, wave/boss/progress
BossFightUIController.cs	UI for boss events and stats
SurvivorShooter.cs	Manages survivors' automatic shooting
Zombie.cs	Zombie health, wave events, respawning
ZombieBoss.cs	Boss logic, input detection, defeat event
BuySurvivorUI.cs	Shop/shop UI, buying/selling survivors
MoneyUIController.cs	Displays player currency
KillsUIController.cs	Displays kill progress per wave
PowerUpButtonUI.cs	UI for powerup pricing/upgrades
LeaderboardManager.cs	Loads and displays leaderboard from DB
SceneNavigation.cs	Handles UI scene switching
UnityMainThreadDispatcher.cs	Ensures Firebase/UI thread safety

SETUP INSTRUCTIONS

B. Key Implementation Details

- GameManager.cs is a singleton. Manages zombie waves, kills, survivor spawns, powerups, boss fight sequence, saving/loading progress.
- AuthUIController.cs splits secure password handling (SHA-256), registration and login coroutines, and scene transitions.
- Firebase functions run on background threads. UI updates or object spawning must happen on the Unity main thread.
- UnityMainThreadDispatcher.cs is used to queue callbacks:

```
UnityMainThreadDispatcher.Instance().Enqueue(() => {  
    // UI update or Unity-specific code here  
});
```

You can insert code excerpts or diagrams for more detail.

See <https://github.com/PimDeWitte/UnityMainThreadDispatcher> for more detailed information

CUSTOMIZATION & EXTENSIONS

5. Customization & Extensions

- To add new survivors:
 - a. Create a new survivor prefab.
 - b. Update GameManager.cs to support the new survivor type.
 - c. Add shop button and cost/UI entry.
- To expand the Firebase database rules or use Firestore, see the full Firebase docs.
- For more input methods, implement additional checks in ZombieBoss.cs or TouchInputManager.cs.

6. Credits

- Game Design & Programming: Mohammad Rashid
- Art/UI: Mohammad Rashid, Goldmetal (Zombies), Aider Art (Soldiers), Guty Kreum (Japanese city background)
- Special Thanks: PimDeWitte for the UnityMainThreadDispatcher, Perplexity AI, Firebase Documentation
- Perplexity AI prompt for this project can be found here:
- <https://www.perplexity.ai/search/feat-gameplay-implement-basic-zNIMBKY9QIm3Da.VNwBorg?25=d>

DOWNLOAD APK

To download the Game directly to your android phone, you can find the APK directly on my Itchio page:

- <https://momora.itch.io/zombiefarm>

Or download from external MEGA link

- https://mega.nz/file/okNA1KBI#y0WHqhP_YgJXLGjXsq9g1Pgo0OB9AGTTiSVPVZfnjOk

Download the APK file

- Scroll to the Download section.
- Click on the ZombieFarmV4.apk file link to start downloading.
- If you see multiple files, download the APK file (ends with .apk).

Enable Installation from Unknown Sources

Since ZombieFarm is not from the Google Play Store, your Android device needs permission to install apps from unknown sources. Steps vary by device and Android version:

- On Android 8.0+ (Oreo and later):

§When you try to install the APK, you should be prompted to allow installation from that source (e.g., your browser or file manager).

Tap Settings in the pop-up and enable it.

- On older Android versions:

oGo to Settings > Security and enable Unknown Sources.

- After enabling, return to installation.

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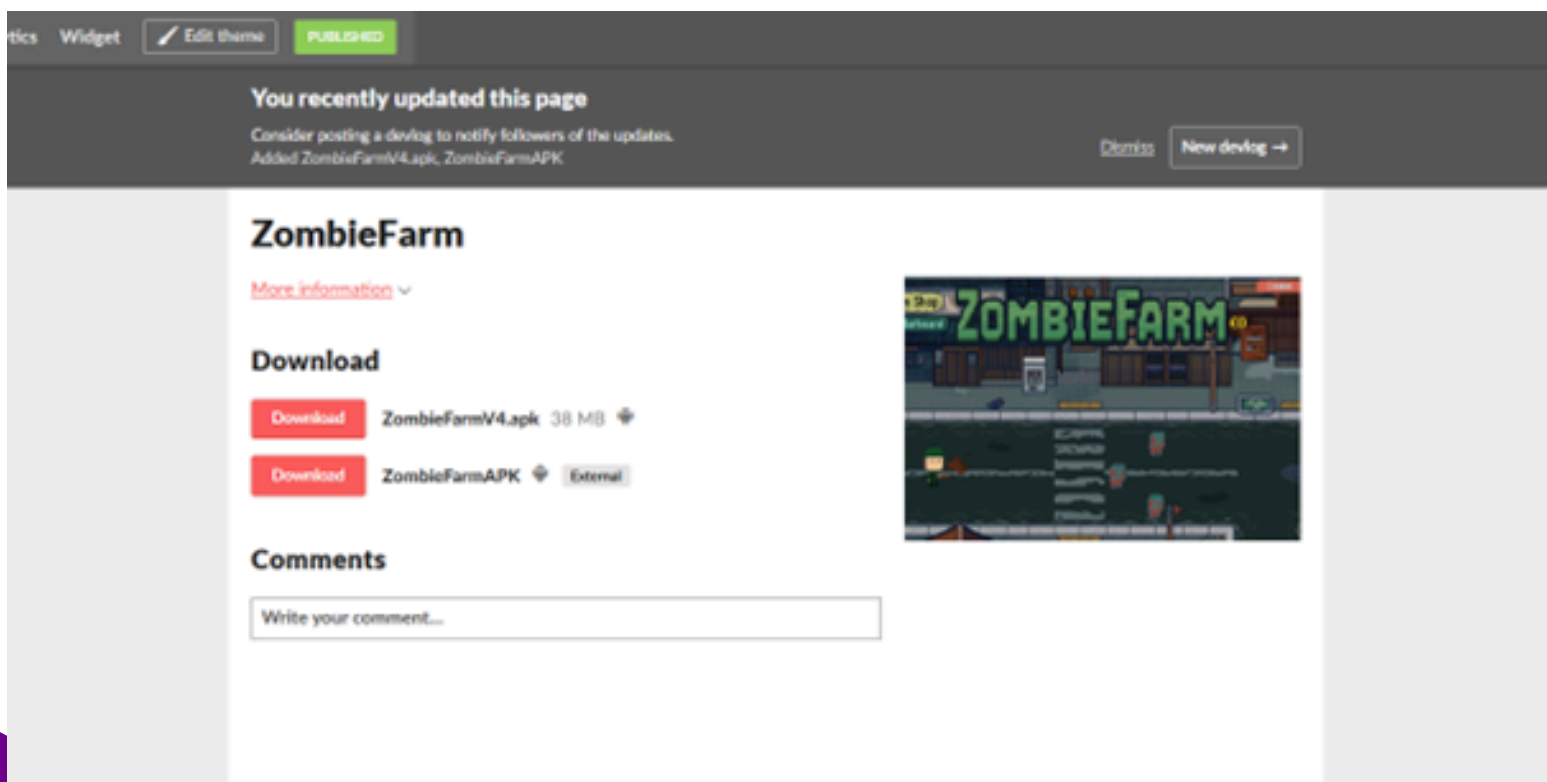
- https://mega.nz/file/okNA1KBI#y0WHqhP_YgJXLGjXsq9g1Pgo0OB9AGTTiSVPVZfnjOk

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- After enabling, return to installation.

Install the APK

- Open your device's file manager or browser downloads.

- Tap the downloaded ZombieFarmV4.apk file.

- Follow the on-screen prompts to install the game.

Launch ZombieFarm

- Once installed, open the app from your app drawer or home screen.

- Enjoy playing!

